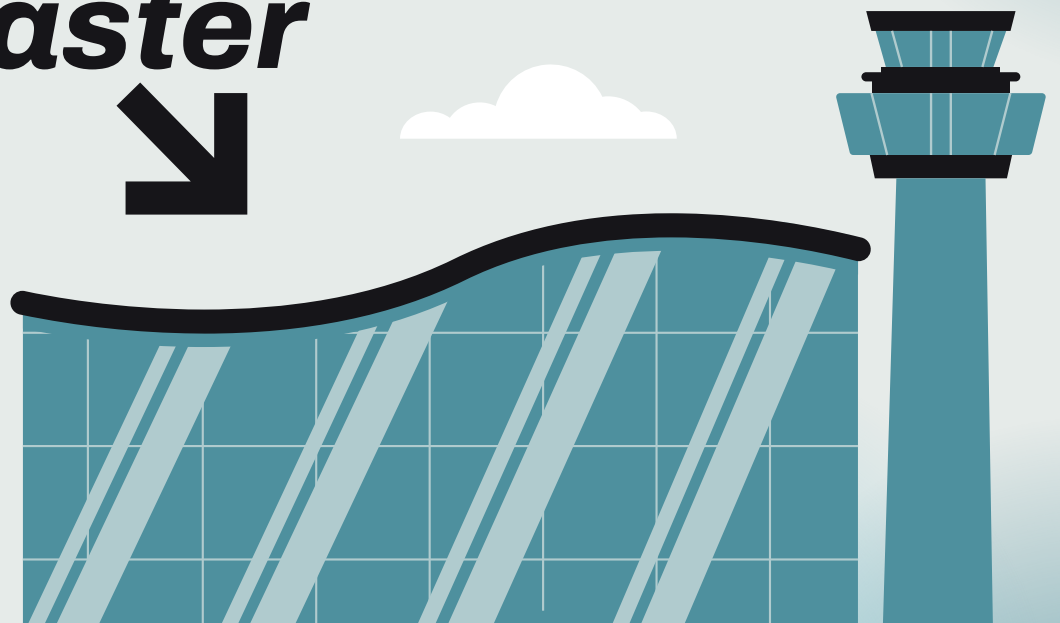


How Media Sentiment Fuels Public Fear of Aviation Disaster

APAN 5205



Research Question

01



What is the correlation between aviation crashes and Google search volumes concerning it, regardless of media coverage?

02



Do increased plane crash headlines correlate with increased Google search volumes related to flight fear?

03



Is media coverage of aviation crashes disproportionately high compared to the genuine number crash occurrences?

04



Does negative sentiment in aviation disaster news headlines predict an increase in public fear of planes?

Data Sources



NewsAPI*

Fetches real-time news articles from multiple sources based on keyword queries.

NewsAPI

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NYT API

Fetches real-time news articles from multiple sources based on keyword queries.

ASN*

A database that provides detailed reports on aviation accidents and safety incidents worldwide.

API (Application Program Interface) ASN (Aviation Safety Network)

Linguistics



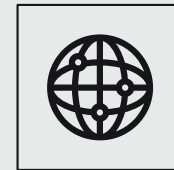
Bing

Classifies words as simply positive or negative to count sentiment polarity.



Affin

Assigns each word a numeric sentiment score -5 to +5



NRC*

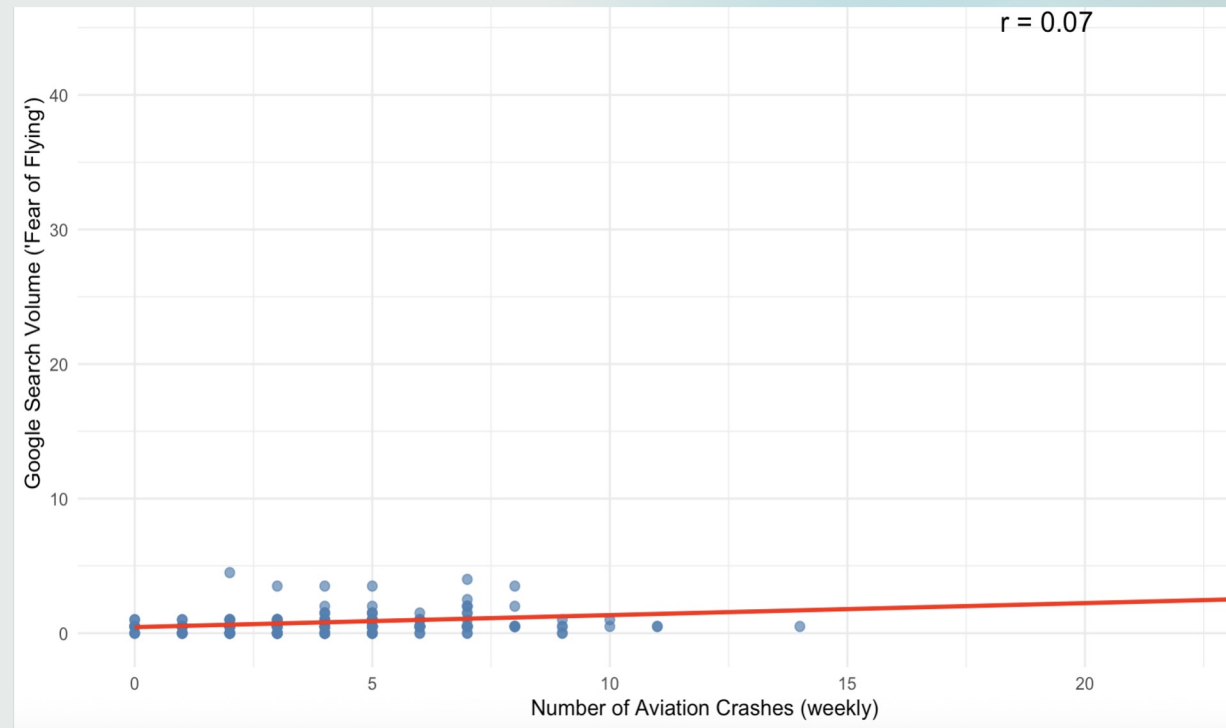
Labels words with emotions to capture nuanced sentiment

✚ ***Correlation between aviation crashes and Google search volumes concerning it, regardless of media coverage***

Pearson Correlation

There's no strong link
between crashes and
search volume

**($r = 0.07$,
 $p = 0.23$,
 $R^2 = 0.0055$).**



✚ *Correlation between aviation crashes and Google search volumes concerning it, regardless of media coverage*

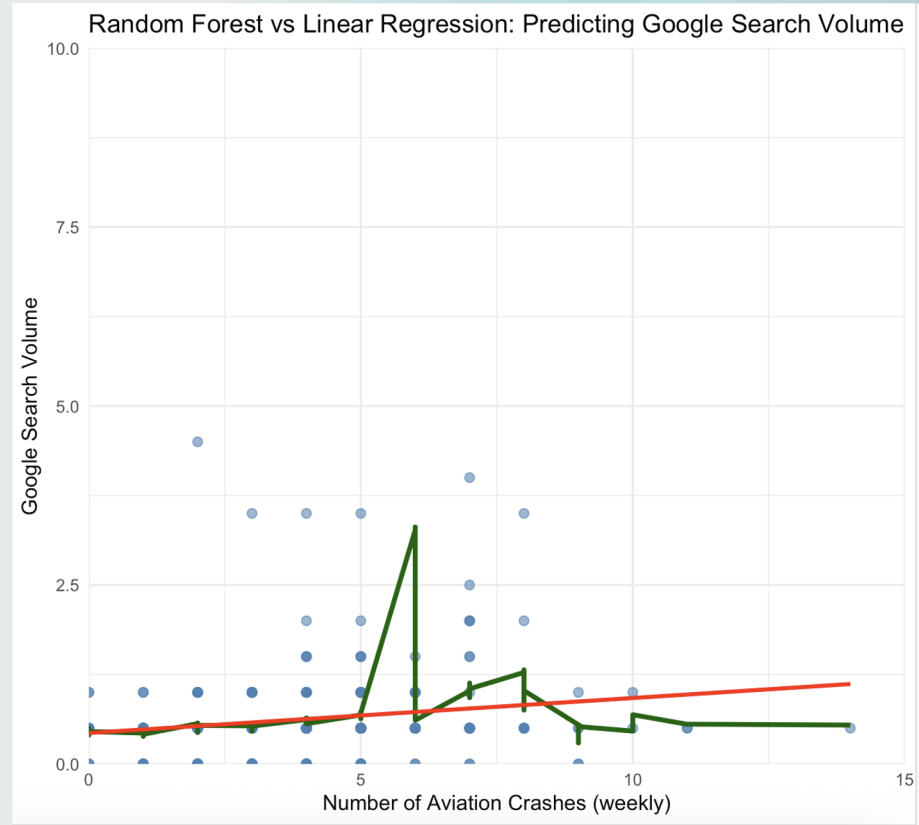
Linear regression & Random Forest

Linear regression shows the relationship is **not significant** — nothing consistent or predictive.

Random Forest picks up a **possible threshold** — search volume only spikes when crash counts hit around 6/week.

So fear doesn't build gradually — it jumps in rare moments.

Overall, search interest moves more than crash numbers, **suggesting external factors mostly drive fear, not just data.**

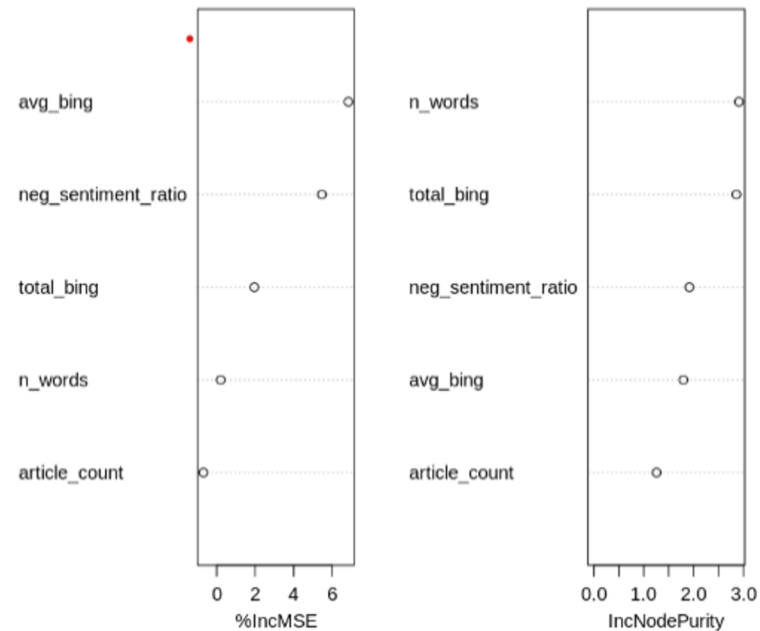


Do increased plane crash headlines correlate with increased Google search volumes related to flight fear?

Random Forest Feature

While features like the number of sentiment words (n_words) and total sentiment (total_bing) show relative importance, none are strong enough to reliably predict fear-related Google searches. This suggests that headline sentiment alone isn't sufficient.

Random Forest Feature Importance (Bing Sentiment Features)



➤ ***Do increased plane crash headlines correlate with increased Google search volumes related to flight fear?***

Random Forest

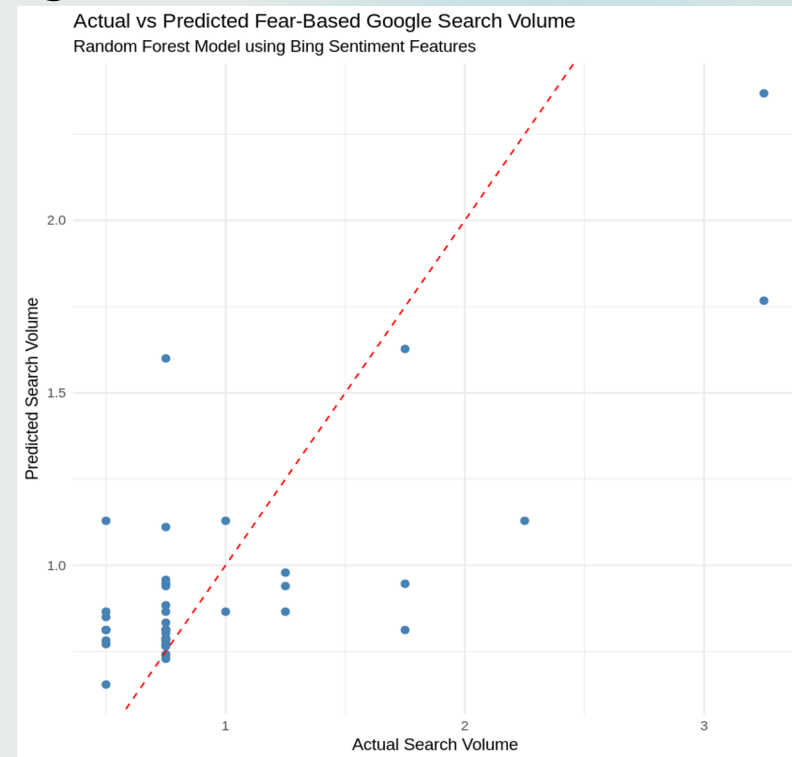
Actual vs. Predicted search volumes show wide dispersion, especially under-prediction for spikes. Some signal exists, but headline sentiment alone doesn't reliably predict fear-based search behavior.

Ntree = 500

Number of variables tried at each split = 1

mean of squared residuals = 0.406612

%Var explained = -11.08



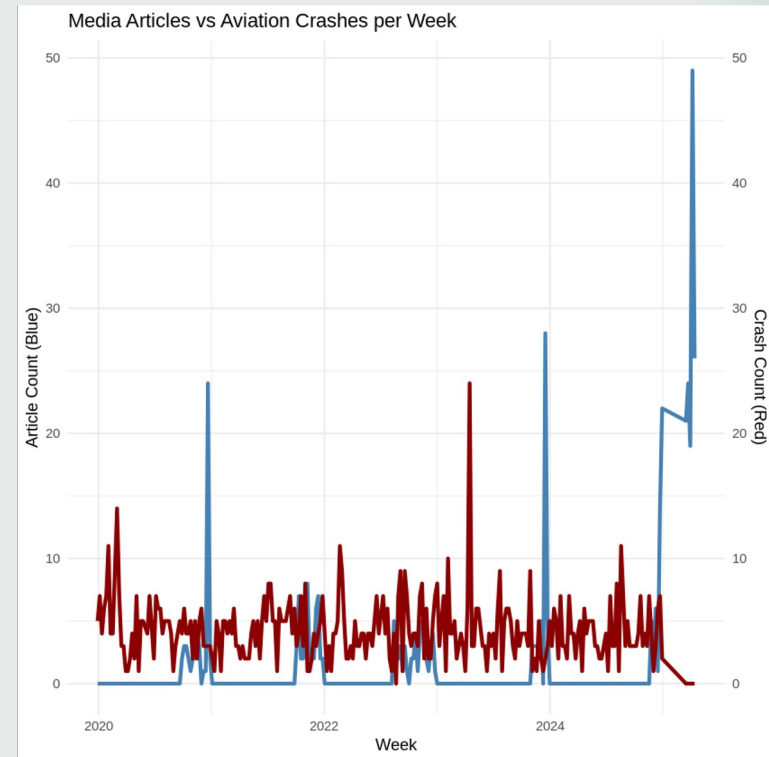
Is media coverage of aviation crashes disproportionately high compared to the genuine number crash occurrences?

Time Series + Ratio-Based Analysis

Key Metric Used: The article-to-crash ratio ($\text{article_count} / \text{crash_count}$) highlights weeks where media coverage is disproportionately high relative to actual crash events.

The average ratio is **0.37**, but the maximum hits **14**.

Shows **distorted public perception** and elevated **fear of flying** during isolated events.



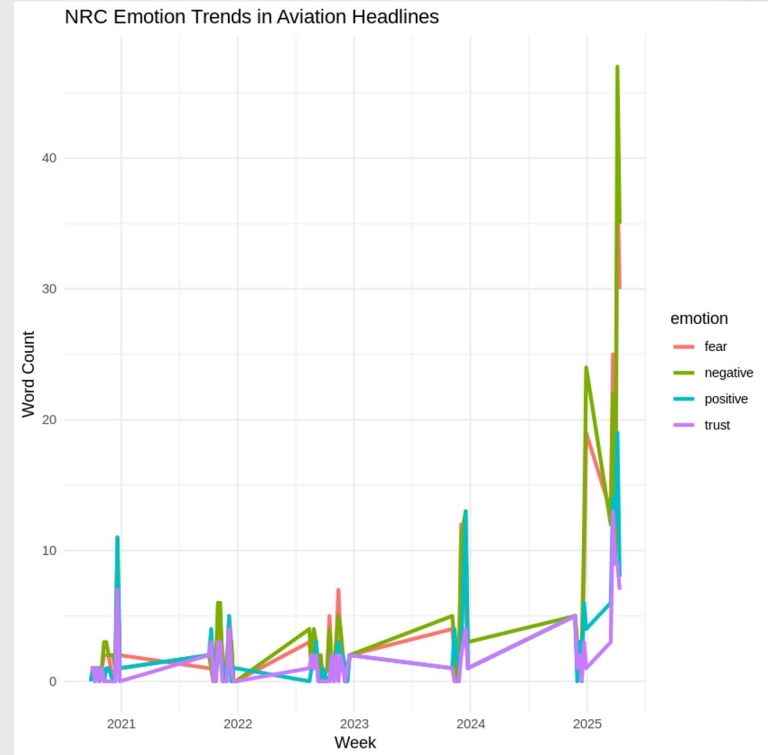
Is media coverage of aviation crashes disproportionately high compared to the genuine number crash occurrences?

NRC Emotion Trends in Aviation Headlines

Tokenized headline words and matched them to NRC emotions, then plotted weekly word counts for key emotions.

Sharp emotional spikes (especially *fear* and *negative*) that coincide with major media surges.

Media overreaction is not just quantitative (volume), but also qualitative (emotional tone).



Is media coverage of aviation crashes disproportionately high compared to the genuine number crash occurrences?

Linear Regression

Model used: A simple linear regression model:
 $\text{avg_hits} \sim \text{afinn_score}$ where:

avg_hits = average weekly Google Trends interest in
“fear of flying”

afinn_score = average sentiment polarity of headlines
that week (AFINN)

Coefficient for afinn_score : -0.0053

$p\text{-value} = 0.939$, indicating no statistical significance
 $R^2 = 0.0002$, meaning AFINN sentiment explains
virtually none of the variance in search behavior



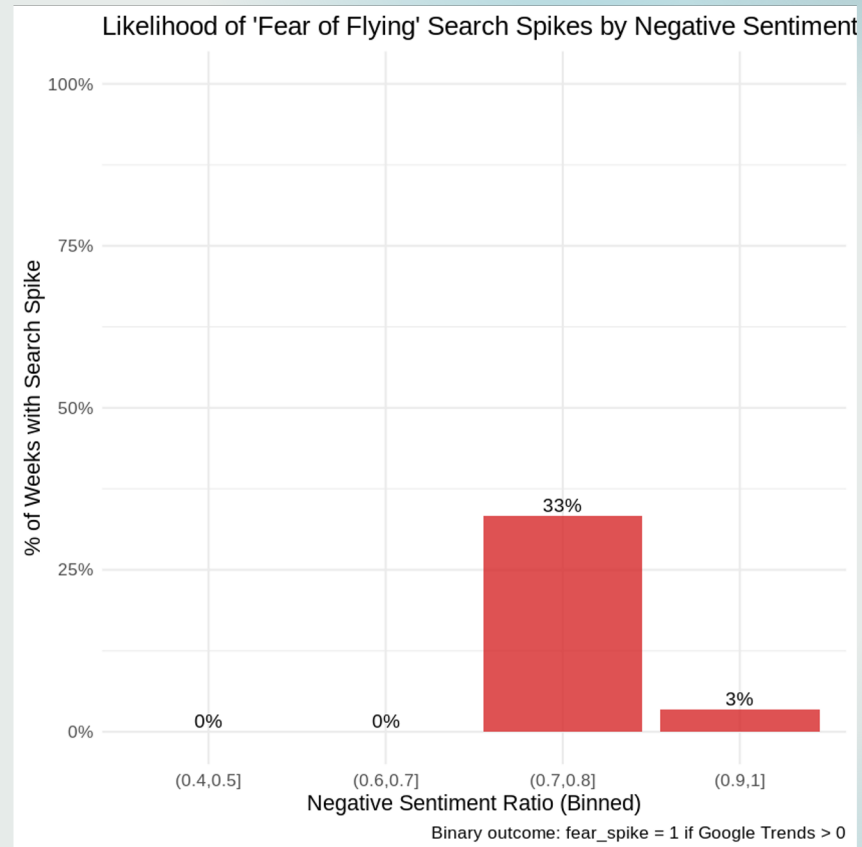
Does negative sentiment in aviation disaster news headlines predict an increase in public fear of planes?

Binning-Based Aggregation with Proportion Comparison

The 33% spike in search activity when headlines are heavily negative

Other bins show 0–3%, reinforcing that public fear is most sensitive to high negativity.

Evidence of a threshold effect — fear spikes only after negative sentiment crosses a certain point



Recommendations & Limitations

Recommendations

Media Editors Should Monitor Negative Sentiment in Headlines

Public Institutions Should Track Search Trends as Fear Signals

Improve Crisis Communication During Aviation Events

Limitations

Correlation $\neq$ Causation

Only Headlines Were Analyzed

Simplified Sentiment Measures

Limited Scope of Google Trends

Just having a headline isn't enough



TEAM MEMBERS



GARIMA GUPTA

“GG2971”



DEPALI KULKARNI

“DVK2109”



SANA KHAN

“SK5528”



**TESS
IBRAHIMCHAH**

“TI2277”

